

A Chatroom, Independent Attempts, and One Agent on Three Tasks

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Abstract

We compare a single agent, a three-seat chatroom, and selection from independent attempts on machine-scored tasks at roughly matched mean spend. All results are dated measurements of `claude-sonnet-5` under the `sonnet` alias on recorded subscription accounts, not estimates for other models or deployments. In Study 1, the chatroom passed both interpreter tasks on every seed, against 61/80 single-agent passes across their shared task family. The vote arm passed 75/80; direct tests did not distinguish it from the chatroom. On a numeric-formatting task, one agent passed 14/40, the chatroom 6/40, and the vote 7/40. The lower chatroom rate was significant under the pre-registered test but not a later Fisher test. Voting selected the common wrong implementation even though correct attempts were often available. The vote prediction succeeded on one interpreter task but failed its stated significance clause on the other.

The same recorded alias and settings yielded sharply different thinking-token distributions across subscription accounts. Study 2 mostly ran on the long-thinking account, leaving its pre-registered short-thinking primary comparisons essentially unobserved. Its small, exploratory long-thinking comparisons distinguished no arm from one agent. These data do not identify the cause of the account difference or the marginal effect of any chatroom mechanic. The comparison shows why serving-account provenance belongs with each run’s outcome.

1 Introduction

A chatroom can give several language-model agents a chance to challenge and revise one another’s work. It also buys more attempts at the task. To assess what the conversation contributes, the relevant comparison is with independent attempts chosen without seeing the answer, at a similar cost. This paper makes that comparison on a small, deliberately difficult task suite, then shows how a change in the serving account limited a later, broader study.

The measured system is `agent-chatroom-mcp`. Every run reported here used `claude-sonnet-5` under the `sonnet` alias between 18 and 20 September 2026. The subscription account active at run time is reconstructed from a sanitized switch log. Pass rates, costs, and which task was near ceiling belong to those dates, model and accounts. They are not predictions for newer models. The study method and the need to retain serving provenance do carry forward; the observed minority-versus-majority pattern is a hypothesis to test elsewhere.

Study 1 used seeds 101–140 on three fixed tasks, comparing one agent (A), a three-seat chatroom (C), and independent attempts with oracle-free selection (K). The two interpreter programs share one scoping trap and count as a single task family. The third task asks for exact numeric formatting. Arms A and C were interleaved by seed; K ran later that day, before the account switch logged at 16:38 UTC. Mean spend was matched between C and K, but individual paired seeds were not cost matched. All outcomes came from a machine oracle, not from the agents’ own verdicts.

Study 2 used seeds 501–520 and added a higher-effort single agent (AH) and a builder–reviewer pair (B). Its pre-registered primary comparison targeted short thinking. The run stopped with missing cells after most work was served by the long-thinking account. The resulting long-thinking comparisons were specified after data had been seen and remain exploratory.

The paper makes three narrow points. In Study 1, direct tests did not distinguish the chatroom from independent selection, even though both could outperform one agent on the interpreter family and both had lower observed success on printf. Across recorded accounts, thinking tokens differed under the same alias and settings, with one long-thinking single agent near ceiling on the measured interpreter task. Study 2 cannot answer its primary question: the intended short-thinking stratum was almost absent. These points are reported with the failed pre-registered vote clause, the test revision, costs, and the missing data, rather than folded into a claim that chatrooms or voting generally help or hurt.

2 Evaluated system

The evaluated build of `agent-chatroom-mcp` gave each chatroom seat an independent opening before shared discussion. Seats could propose and amend one answer, challenge a quoted span, and vote on the current proposal. Agreement votes had to quote the proposal; the room used a supermajority rule that required unanimity with three active seats. A verification gate required a machine-parseable verdict from a connection other than the proposer, and the hub assigned that reviewer. The verify gate and reviewer assignment were present before both studies. Kick, replace and heartbeat changes dated 22 September were later additions and were not evaluated here. Study 1 chatroom cells record hub entry hash `66f9d135...`; run artifacts carry the revision used for each cell. The design offers routes for correction, but this paper does not isolate any route’s effect.

This distinction between a build’s presence and its measured effect is central to the system description. The revision hash establishes which code the chatroom cells entered; it does not say which part of that code determined an answer. A later hub feature cannot be credited with an earlier result merely because it now appears in the repository. The outcome comparison is between complete arm protocols as recorded, with any finer process account left to the limited seat summaries.

Arm A used one agent on the same public task brief. Arm K ran fresh, non-communicating A-style attempts and selected one answer without access to the hidden oracle. For the two interpreter tasks, it chose the most frequent normalized final answer, breaking a tie by sub-seed. For printf, it ran candidate programs on a fixed public-specification probe set and chose the implementation with the greatest execution agreement. Arm B in Study 2 passed one builder’s output to a reviewer for possible revision; AH used one agent at higher effort. Thus B was run, while Study 1 has no B cell.

All cost-per-correct figures use API list prices and include the attempts needed to produce the selected answer. They describe these accounts and dates. A chatroom conclusion, a vote, or a participant’s reported confidence never served as the correctness oracle. This matters particularly for process evidence: the retained Study 1 room summaries can say that a correction happened, but cannot prove that a challenge, vote, or verification gate caused it.

3 Method

Tasks and scoring. Study 1’s tasks were admitted after a single-agent pilot screened about 45 designs for a discriminating pass rate. The admitted tasks each contain one known trap.

`stamp-interpreter` and `stamp-2` ask for programs implementing the same invented-language scoping rule; they are two fixed instances of one family, not independent evidence from two domains. `bench-printf-format` asks for exact formatting where a natural implementation using the host runtime’s shortest representation disagrees with the specified output. The same hidden, machine-checked fixtures scored every arm. The seeds vary attempts on fixed task instances, rather than sampling new tasks. The task suite therefore supports comparisons on these traps, not an estimate across software engineering work.

The pilot admission rule retained tasks whose single-agent rate on ten seeds fell between 0.3 and 0.7, inclusive. That rule selected tasks where a comparison had room to show a gain or a loss, but it also selected for particular model weaknesses. The scope-capture tasks differ in their surface programs, yet share the same rule about access to a caller’s local variables. Treating their outcomes as separate families would count the same failure mode twice. The `printf` task supplies a different mechanism: a program can look plausible and pass its own checks while still mishandling the exact representation required by the hidden fixture.

Study 1 design. For each task, A and C ran on seeds 101–140, interleaved per seed. A’s budget ceiling was tied to C’s realized spend; matching did not force the same realized spend. K was separately pre-registered after A and C results but before K ran, on the same seeds and tasks. Its number of attempts was chosen from the earlier mean A and C costs, with each attempt run afresh. The selector had no oracle input. The fixed K group sizes were ten attempts for `stamp-interpreter`, eight for `stamp-2`, and seven for `printf`. The per-attempt cap left room for an ordinary A-style attempt; total group cost included every attempt even if it was killed or not selected. The matching target was mean realized C spend, not identical spend on every seed. The single-agent baseline likewise had a ceiling rather than a requirement to consume the chatroom’s full budget.

K ran from 13:22 to 15:33 UTC, before the account switch recorded at 16:38 UTC; an earlier shift-time estimate was superseded by that switch log. The comparison between K and C still spans separate collection windows. K exceeded its paired C spend on 19, 10, and 9 of the 40 seeds for the three tasks, despite matching at the mean.

The original A–C protocol specified a two-proportion z-test, using Fisher’s exact test for small cells, separately by task. After the data, the two interpreter tasks were pooled as a family and A–C was retested with Fisher’s exact test and Holm correction across the two families. We report that revised family result alongside the original per-task tests. The K pre-registration fixed six two-sided Fisher comparisons, K against A and C on each task, with Holm correction across all six. Pooled K comparisons on the interpreter family are exploratory and do not replace those registered tests. A result that fails to distinguish two arms is not an equivalence test.

The reported p -values are tests of observed pass counts, with the comparison families specified above. The common seed labels put arms on the same task brief, but the attempts are separate model runs; the Fisher calculations do not treat them as paired observations. In Study 1, statistical significance is therefore reported at the resolution of the tested task or family, not as evidence that a particular room action worked. The direct K–C comparisons answer a different question from separate K–A and C–A decisions.

Study 2 design and missingness. Study 2 specified five arms on seeds 501–520: A, AH, B, K and C. Its primary tests used the short-thinking stratum. The account switcher was not known to the harness at launch. A sanitized log later associated each run with its active account; mixed-account seeds were separated from within-regime comparisons. The run stopped at 190/200 cells, leaving ten absent. `Printf` seed 501 is kept in descriptive totals but excluded from inference after

Table 1: Study 1, A versus C by task family. The family pooling and Fisher test were post-data revisions; the pre-registered per-task tests are described in the text.

Family	A pass/ n	C pass/ n	Fisher p (raw)	Holm p ($m = 2$)	Sig.
interpreter (stamp-interpreter + stamp-2)	61/80	80/80	0.000001	0.000002	yes
printf (bench-printf-format)	14/40	6/40	0.069165	0.069165	no

a documented cap deviation. One printf chatroom run has unknown cost. The long-thinking tests were added after much of that stratum had been observed, and use 19 seeds per arm on **stamp-interpreter** and 15 on printf. They use Fisher tests with Holm correction across the registered comparison family, but their timing makes them exploratory. An A-arm thinking-token sentinel stratifies the report; it cannot rule out a change later in the same seed.

Account measurement. The account analysis joins single-seat run intervals to switch times and uses only runs with thinking telemetry for the thinking-token comparison. Long thinking was defined in the Study 2 pre-registration as at least 4,000 thinking tokens. Five of 1,947 eligible runs straddled a switch and were excluded. Output-token distributions give a second descriptive account comparison, but output across a three-seat room is not directly comparable to a single seat. Neither the active account nor effort was randomized. The client records the association, not the server-side reason for it.

4 Results

4.1 Study 1: conversation and independent selection

Table 1 gives the A–C family comparison. Across the two interpreter programs, A passed 61/80 and C passed 80/80. The family-pooled Fisher test, adopted after the runs, gives $p = 0.000001$ before Holm adjustment. The original pre-registered per-task tests also rejected A–C on each interpreter program; the pooled result is a revised analysis, not the pre-registered test. K passed 36/40 on **stamp-interpreter** and 39/40 on **stamp-2**, or 75/80 pooled. Its pooled comparison with A ($p = 0.0033$) is exploratory. The registered K tests show a narrower result: K beat A on **stamp-2** (Holm $p = 0.009$), but was not distinguished from A on **stamp-interpreter** (Holm $p = 1.000$). The pre-registered prediction said K would beat A on both. That clause failed.

This is not repaired by pooling the two programs after the result: the K prediction explicitly made a claim about each program. The observed K rate on **stamp-interpreter** was higher than A’s, but the registered comparison could not distinguish them. By contrast, the direct K–C test asks whether the two aggregation arms differ. Keeping those tests separate avoids treating a missed threshold against A as proof of a chatroom-specific advantage.

Direct K–C tests did not distinguish the arms on any Study 1 task: every Holm-adjusted p was at least 0.578. Pooling the interpreter tasks gives 75/80 for K and 80/80 for C, with an exploratory Fisher $p = 0.059$. Neither result establishes equality. In particular, a significant C–A result and a nonsignificant K–A result on **stamp-interpreter** are not themselves a test of C against K. The direct comparison is the relevant test, and the available sample leaves a small difference unresolved.

On **bench-printf-format**, A passed 14/40, C 6/40, and K 7/40. The pre-registered A–C z-test gave $p = 0.039$; the later family-based Fisher test gave $p = 0.069$, so the significance decision depends on the stated test. K was not distinguished from A (Holm $p = 0.578$) or C. The observed

rates on this task are lower for both aggregation arms; they do not support a general assertion that either method lowers success on other tasks.

The independent-attempt evidence explains K’s printf outcome more specifically. In a fixed pool of 40 A attempts, all 14 passing implementations agreed on every one of 9,650 public-specification probes, and all 26 failing implementations also agreed with one another. Thus execution agreement identified the larger wrong cluster. In the real K groups, at least one correct attempt was available in 37/40, but the oracle-free selector submitted a correct one in only 7/40. The availability count is an oracle ceiling, not a deployable arm. The C artifacts retain pass or fail, but not the failing implementations; the same modal-error explanation for C is an association, not an observed C selection mechanism.

The interpreter answers show the opposite population pattern in Study 1. For **stamp-interpreter**, the seven A failures among 40 were identical; for **stamp-2**, 11 of the 12 failures were identical and one differed. Here the correct answer was the majority of A answers in the observed pool. The room summaries add diagnostic context: at least 26 of 40 **stamp-interpreter** rooms and 14 of 40 **stamp-2** rooms reported an erroneous opening followed by another seat’s correction. These are manual lower bounds from final seat summaries, without retained raw room events. They do not identify a causal chatroom mechanic.

The K failures also show why a favorable population distribution is not a guarantee for one vote group. A correct answer can be common over all A attempts yet lose within a particular set of fresh attempts. On **stamp-interpreter**, some failed K groups had a wrong-answer plurality and others were ties resolved against the correct answer. The fixed tie rule made those losses reproducible; it did not consult the oracle. The retained C summaries cannot be placed on the same group-vote scale.

The extra computation had a substantial price in these runs. C’s cost per correct answer relative to A was 8.56 times on **stamp-interpreter**, 6.16 times on **stamp-2**, and 18.17 times on printf. K and C cost respectively \$0.637 and \$0.602 per correct answer on **stamp-interpreter**, \$0.545 and \$0.595 on **stamp-2**, and \$3.87 and \$4.97 on printf. These are list-price calculations under the recorded accounts and include all K attempts, including those not selected. Mean budget matching did not remove per-seed spend differences, and K’s later collection window remains a design limitation even though the logged account switch followed it.

4.2 Account-associated thinking

Table 2 joins single-seat run times to the sanitized switch log. Among runs with thinking telemetry, account 4 had 384/385 at or above 4,000 thinking tokens. Accounts 1 and 3 had none among 51 combined runs. Median thinking was 7,709 tokens on account 4, against 968 and 716 on accounts 1 and 3. Within each measured task family, the output-token ranges also separated by account. These are strong associations under the same alias and settings, not evidence that the client can identify what the provider changed.

The association matters for interpreting accuracy. Short-thinking single-agent measurements on **stamp-interpreter** came from several cohorts: the admission pilot, the Study 1 A grid, constituent K attempts, a cap control, and the window control. Their estimates ranged from 0.475 to 0.825. The Study 1 grid’s 33/40 was therefore one estimate within an observed span, not a stable model parameter. The separate long-thinking window-control cohort had 78/80 successful constituent single attempts. Study 2’s long-thinking A arm passed 18/19 on that task and cost \$0.136 per correct answer. These are different cohorts, not a randomized account comparison; their gap does not show that either account or effort caused an accuracy change. The source of the variation among short-thinking estimates is also unidentified. This matters when reading a point forecast

Table 2: Recorded subscription account and run-level thinking or output tokens. Generated from the account-switch log and committed run artifacts.

Active account	Runs	Long-thinking runs	Median thinking tokens	Min	Max
Account 1	40	0/40	968	463	1295
Account 3	11	0/11	716	595	1316
Account 4	385	384/385	7709	3742	31604

Task family	Active account	Runs	Median output tokens	Min	Max
bench-printf-format	Account 1	337	4645	3789	6595
bench-printf-format	Account 3	26	4225	3726	5181
bench-printf-format	Account 4	152	25144	16212	43700
stamp-interpreter	Account 1	590	1716	1357	2232
stamp-interpreter	Account 4	313	7310	4494	16896

Single-seat runs (arm A, arm AH, and the constituent attempts of arm K) recorded in `bench/results`, joined by start time to a sanitized extract of the maintainer’s account-switcher log (`bench/results/account-switch-log/switches.json`: switch times and account slot numbers only). 5 of 1947 runs straddle a switch and are left out. Top: runs that recorded thinking tokens; long thinking means at least 4000 thinking tokens, the threshold frozen in `paper/prereg-confirmatory.md`. Bottom: output tokens, which every run recorded, for the two confirmatory task families. Account numbers are slots in the switcher, not identifiers.

Table 3: Study 2 descriptive outcomes by task, stratum and arm. Inference excludes mixed-account seeds and printf seed 501; the eligible long-thinking comparisons use fewer seeds than some descriptive rows.

Descriptive rows retain printf seed 501; it is excluded from both primary and exploratory comparisons.

Family	Stratum	A	AH	B	K	C
stamp-interpreter	long thinking	18/19	19/19	19/19	19/19	19/19
stamp-interpreter	mixed accounts	1/1	1/1	1/1	1/1	1/1
bench-printf-format	short thinking	0/1	0/1	1/1	0/1	0/1
bench-printf-format	long thinking	12/16	12/16	11/16	13/16	10/16
bench-printf-format	mixed accounts	0/1	0/1	1/1	1/1	0/1

for independent voting based on an earlier A grid: a forecast that treats that grid’s rate as fixed inherits its sampling variation. **stamp-2** was not measured under long thinking, so the long-thinking interpreter result is confined to **stamp-interpreter**.

4.3 Study 2: the primary stratum was nearly absent

Study 2’s pre-registered primary used short-thinking seeds. It obtained no eligible short-thinking **stamp-interpreter** seed and only one printf seed. The interpreter’s apparent short-thinking seed crossed accounts between arms and was excluded from within-regime inference. One printf seed also crossed accounts. Table 3 retains those rows descriptively; the printf seed with a cap deviation is likewise descriptive only. The missing ten cells and excluded seeds leave the registered primary comparisons essentially unobserved.

The descriptive table deliberately keeps the mixed and excluded outcomes visible. It is not an inferential sample: a row can contain a run that the registered comparison must omit. In particular, the long-thinking printf descriptive rows retain seed 501, while the long-thinking tests use one fewer seed after that cap deviation. A missing result is not a failed task attempt, and a run with unknown cost cannot be priced as though it cost zero.

Within the exploratory long-thinking stratum, the interpreter comparisons use 19 seeds per arm. A passed 18/19; AH, B, K and C each passed 19/19. On printf, the inferential sample has 15 seeds per arm: A and AH passed 11/15, B and C 10/15, and K 12/15. Every Holm-adjusted comparison against A had $p = 1.000$. That test result means no arm was distinguished in these small samples, not that their effects are equal or that the primary prediction was confirmed. On the interpreter, B cost \$0.269 per correct answer and C \$1.407, with both passing 19/19. The pair’s observed cost was about a fifth of the chatroom’s. The pre-registered cost prediction concerned short thinking; this exploratory long-thinking inequality does not confirm it.

The five-arm run therefore measures a different serving condition from its intended primary stratum. Its result is a bounded description of those recorded long-thinking cells. Recording the active serving account at each run, rather than inferring a uniform condition from the model alias, is necessary to know which comparison was actually made.

5 What this does not show

The three fixed tasks were chosen for known, single-trap failure modes after pilot screening. Repeated seeds measure variation on those task instances; the two **stamp** programs are one family. Their observed minority-versus-majority answer pattern is a specific diagnostic for the vote selector, not a rule for multi-agent systems. Study 1’s direct C–K tests lack the precision to establish equal accuracy. The planned vote-versus-chatroom comparison under one stable serving condition was never run at adequate power.

The different Study 1 collection windows are relevant even though the logged account switch came after K: clock time alone does not guarantee every unrecorded serving detail was fixed. The evidence warrants a direct comparison of the observed pass counts, with uncertainty, rather than a claim that the two protocols were exposed to identical hidden conditions. Conversely, the superseded early shift estimate is not evidence that K crossed the logged account boundary.

The maintainer’s working pipeline—prompt, prior-art agent, plan, build, and devkit review—has never been compared with one agent, a builder–reviewer pair, or a room. The marginal value of review by a non-author was not identified either. B ran in Study 2 and in a small pilot, but the Study 2 primary was essentially absent. A separate build-suite diagnostic reached a 9/9 ceiling in every cell. The Study 1 seat-summary tallies report corrections after disagreement; they are self-reports and cannot separate review from other room activity.

The bundled chatroom build was compared with other arms, but its individual mechanics were not ablated. Regression tests show that gates and assignments operate as coded; one bundled-build trial is not a causal estimate for a gate. The retained room artifacts also lack the event sequence needed to reconstruct which action corrected an answer. No effect size is assigned here to blind openings, challenge, amended proposals, quote-checked votes, reviewer assignment, or verification.

Other engineering observations are outside the task-success comparison. Held waits were measured at 115.6 million versus 53.6 million tokens in one room each on different tasks, so that contrast is observational. OpenRouter sticky-session caching was mock-tested only. Checkpoint trimming has no answer-quality test. None of these results supports a claim about the measured A, B, C, or K accuracy differences.

The active account and effort settings were not randomized. The account association does not reveal whether a plan tier, provider-side experiment, or another serving condition produced the different thinking-token distributions. The measured pass rates, costs, and ceiling on **stamp-interpreter** do not transfer to newer models. What can transfer is the comparison method and the practice of recording model, account, time, and per-run telemetry together. The observed

answer-cluster pattern remains a hypothesis for a new prospective test.

6 Related work

Multi-agent debate provides the immediate setting for this comparison. Du et al. [1] study debate as a route to better answers, while Smit et al. [5] examine how debate strategies compare with simpler baselines. Those papers motivate asking whether a conversational protocol earns its additional cost. Our measured question is narrower: on three fixed machine-scored tasks, did this recorded chatroom differ from a vote over fresh attempts? The results do not adjudicate debate on other tasks or models.

Arm K’s interpreter selector is a form of self-consistency: independently generated answers are normalized and the most frequent is submitted [7]. Its code selector instead uses execution agreement among candidate programs, following the type of signal studied by Shi et al. [4]. In the observed printf pool, agreement favored the common wrong implementation. That is why the candidate-availability ceiling and the submitted-answer rate must be reported separately. The result describes the selector as used here; it is not a comparison among alternative selectors.

This distinction also separates two possible questions about sampling. One asks whether a correct candidate appears anywhere in the set; the other asks whether a rule that cannot see the oracle submits it. The printf groups give different answers to those questions. A larger attempt budget can raise availability while agreement-based selection still follows the modal wrong behavior. The paper reports both quantities because treating availability as the actual K result would overstate what the procedure delivered.

The chatroom’s independent openings, challenge, proposal and verification steps make it tempting to attribute corrected answers to a particular interaction. Work on divergent thinking and premature consensus [3; 8] gives reasons to inspect those interactions. The retained seat summaries provide only lower-bound reports of correction. Without an ablation or structured event trace, the present paper cannot test a causal account of those mechanics. Work on minority answers in multi-agent debate [2] likewise frames the wrong-cluster observation as a testable question rather than an established general rule.

Finally, the hidden machine oracle keeps the agents’ own consensus separate from correctness. Research on language-model judges [9; 6] concerns a different measurement route; none of the pass rates here uses such a judge. Our main measurement complication arose earlier, at model serving: the same recorded alias and settings accompanied different thinking-token distributions on different active accounts. The account log and token telemetry therefore belong alongside accuracy and cost in any attempt to reproduce these particular comparisons.

A Reproducibility

The repository’s `paper/README.md` gives the commands for inspecting the five-arm snapshot, regenerating tables and figures, auditing prose numbers, and building the manuscript. The two study specifications are `paper/protocol.md`, `paper/prereg-arm-k.md`, and `paper/prereg-confirmatory.md`; dated deviations and exclusions are in `paper/amendments.md`. Those records distinguish a planned comparison from a later analysis.

For Study 1, the generated A–C family table is a post-data family analysis, while the original per-task tests remain in the suite report. The arm-K report keeps its pre-registered six-test Holm family; the additional pooled interpreter numbers are explicitly exploratory. For Study 2, the generated

report prints descriptive cells and the inferential samples separately. The latter remove the mixed-account seeds and printf seed 501, and distinguish the registered short-thinking primary from the later long-thinking tests. Reading one descriptive denominator as an inferential denominator changes the question.

The outcome artifacts are under `bench/results/rq1-suite`, `bench/results/rq1-arm-k`, and `bench/results/rq1-confirmatory`. The sanitized account switch times are in `bench/results/account-switch-log/switches.json`. Generated reports in `paper/generated` retain the per-arm denominators, prices, strata, and warnings used here. Study 1’s account-window qualification follows the logged switch time; the earlier local-time estimate in the amendments was superseded. For the room-correction tallies, `docs/arm-c-process-trace.md` states the manual criteria and names representative artifacts.

The manuscript inputs generated tables without editing them. `paper/regen-manifest.sh` names the generators and their committed inputs; `scripts/paper-number-audit.ts` checks every typed decimal, fraction, and percentage in the section sources against those inputs and records. The audit confirms provenance of a printed number, not its interpretation. Its source list includes the manual process-trace note only for the reported lower-bound room tallies. The cost-table generator also places the K collection window before the switch time from the logged record; regenerated table rows were not edited to make that correction. Re-running the five-arm report should preserve its missing, mixed-account, and excluded rows rather than silently filling or pooling them.

Author and AI disclosure

Author. [Maintainer name, affiliation, and contact to be filled before submission.]

AI disclosure. Language-model agents helped build the evaluated system, run its recorded experiments, and draft this manuscript. The human author is responsible for checking the evidence, citations, and final submitted text. No agent is listed as an author.

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